AWS AI PRACTITIONER PROJECT

Hands-on AWS Services & AI Evidence Documentation

1. Project Overview

This document presents screenshot-based evidence of hands-on activities completed using AWS services. The project covers cloud compute, object storage, analytics, content delivery, generative AI, voice application testing, natural-language processing and computer vision.

2. Objectives

- Understand practical AWS service workflows through the AWS Management Console.
- Explore compute, storage, analytics and content delivery.
- Experiment with AWS AI services including Bedrock, Comprehend and Rekognition.
- Organize hands-on evidence for a GitHub portfolio.

3. Services Covered

#	Service / Activity	Evidence
1	HTML/CSS Web Interface	Navigation bar and dashboard menu
2	Amazon S3	Lifecycle transitions and expiration
3	Amazon Athena	E-commerce data query results
4	Amazon CloudFront	Hosted website
5	Amazon EC2	Instance management
6	Alexa Developer Console	Skill simulator testing
7	Amazon Bedrock	Generative AI testing
8	Amazon Rekognition	Text detection
9	Amazon Comprehend	Key-phrase analysis
10	Amazon Rekognition	Image properties

4. Screenshot Evidence

Web Navigation Bar

A simple web interface containing Home, About, Services and Contact navigation items and a dashboard menu.

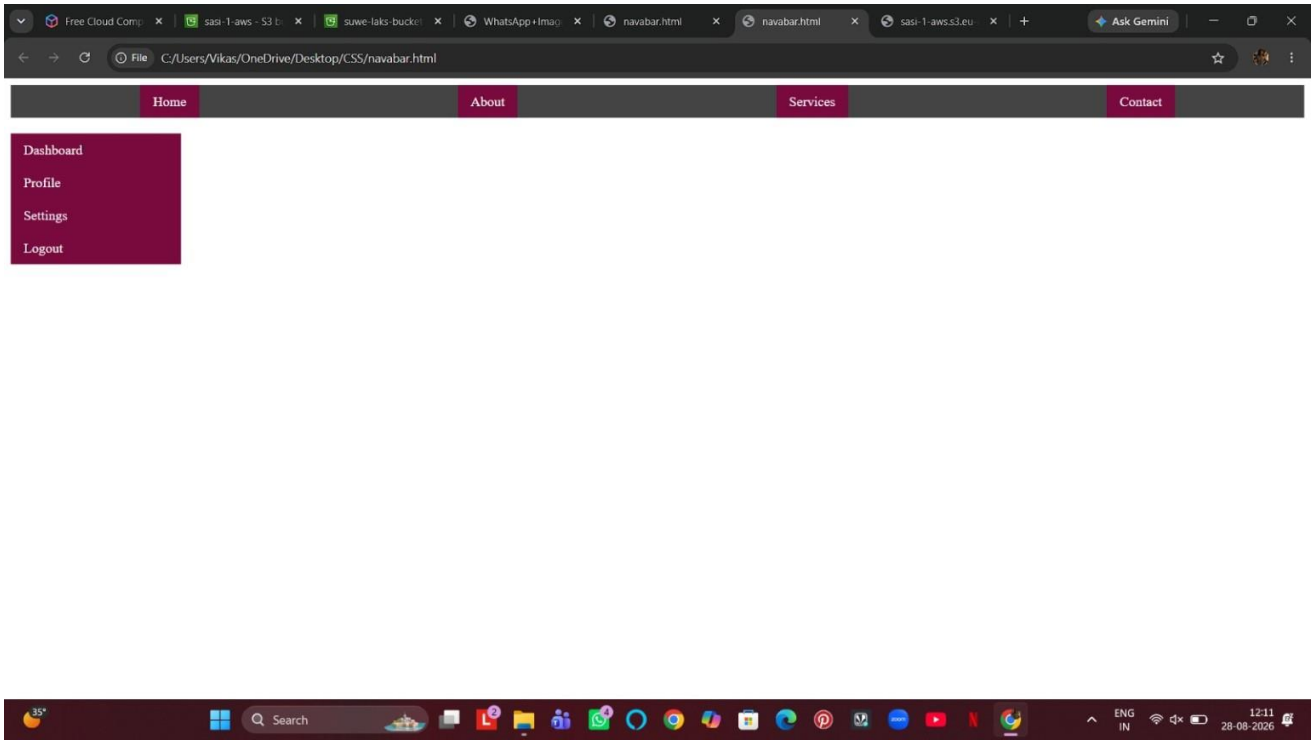


Figure 1: Web Navigation Bar

Amazon S3 Lifecycle Configuration

Shows automated storage lifecycle transitions from uploaded objects to Standard-IA, then One Zone-IA, followed by expiration.

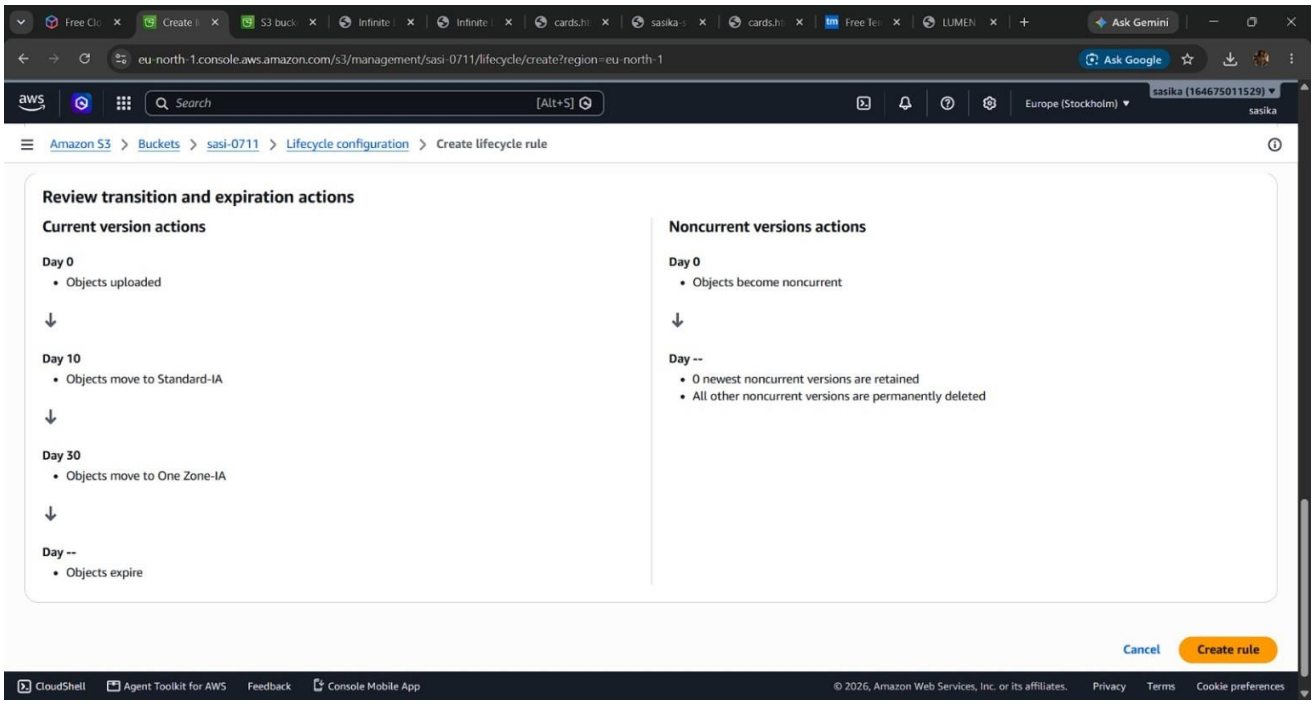


Figure 2: Amazon S3 Lifecycle Configuration

Amazon Athena Query Results

Shows an Athena query result table containing e-commerce/order fields such as order ID, customer ID, product ID, quantity and discount.

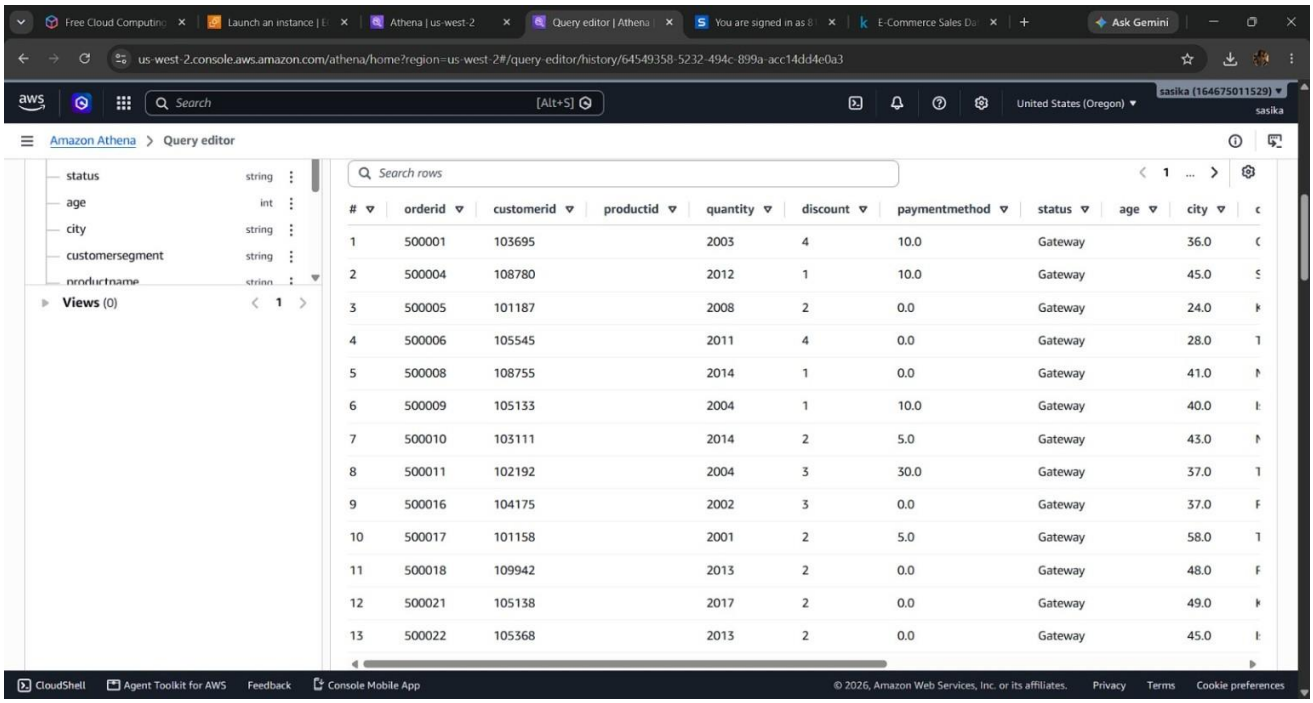


Figure 3: Amazon Athena Query Results

Amazon CloudFront Hosted Website

Shows a website being accessed through a CloudFront distribution.

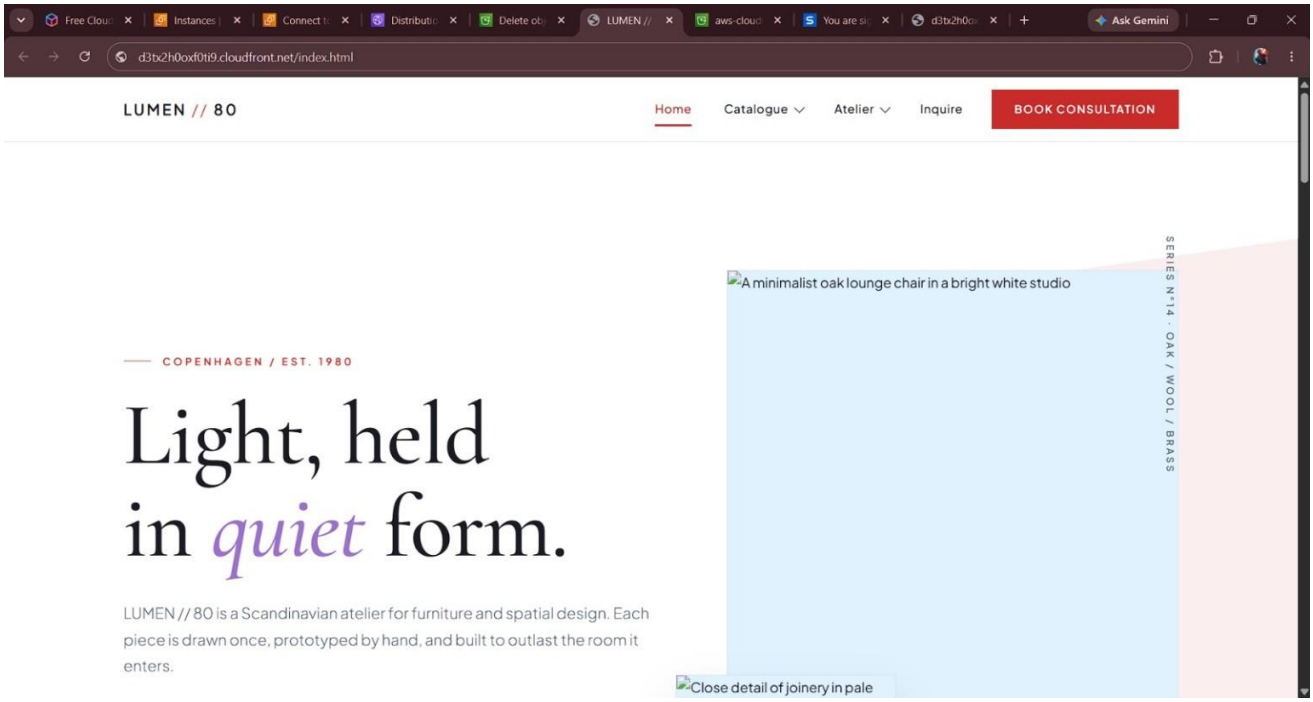


Figure 4: Amazon CloudFront Hosted Website

Amazon EC2 Instances

Shows EC2 instances with running/terminated states, instance types, availability zones and status checks.

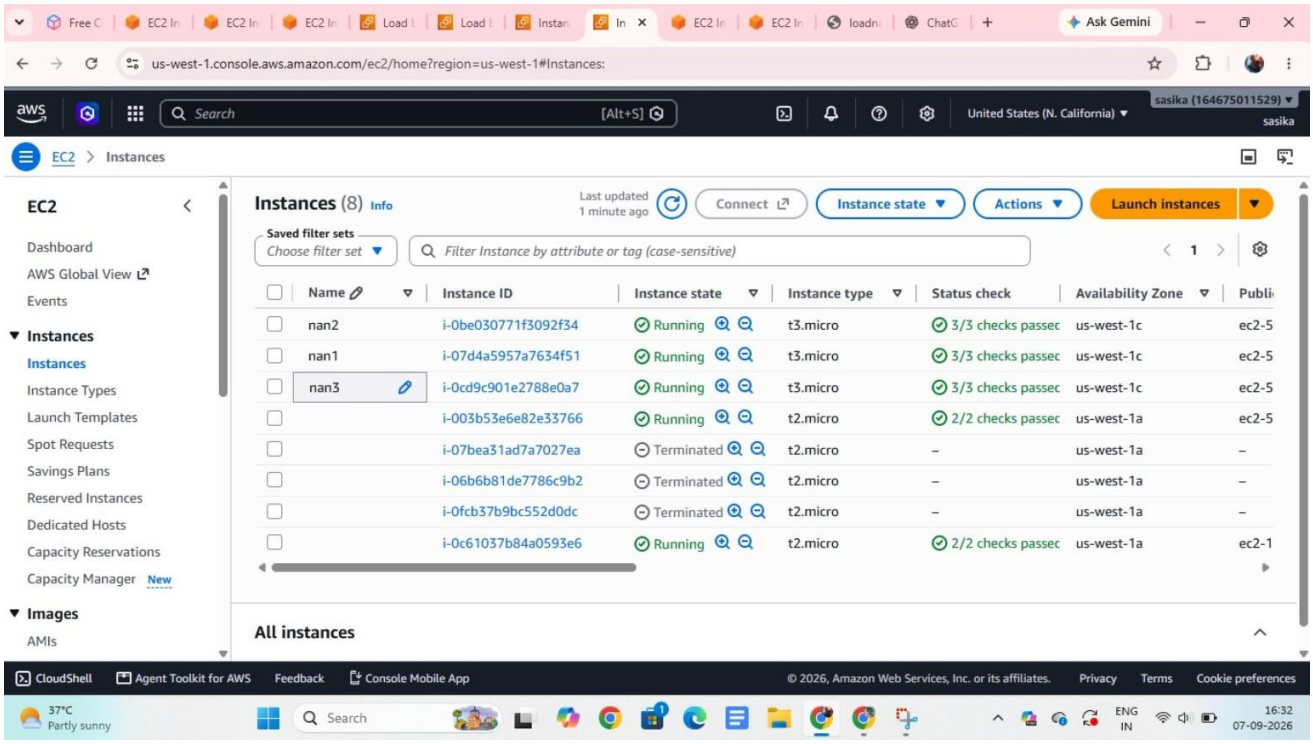


Figure 5: Amazon EC2 Instances

Alexa Developer Console

Shows Alexa skill testing using the simulator and an example response.

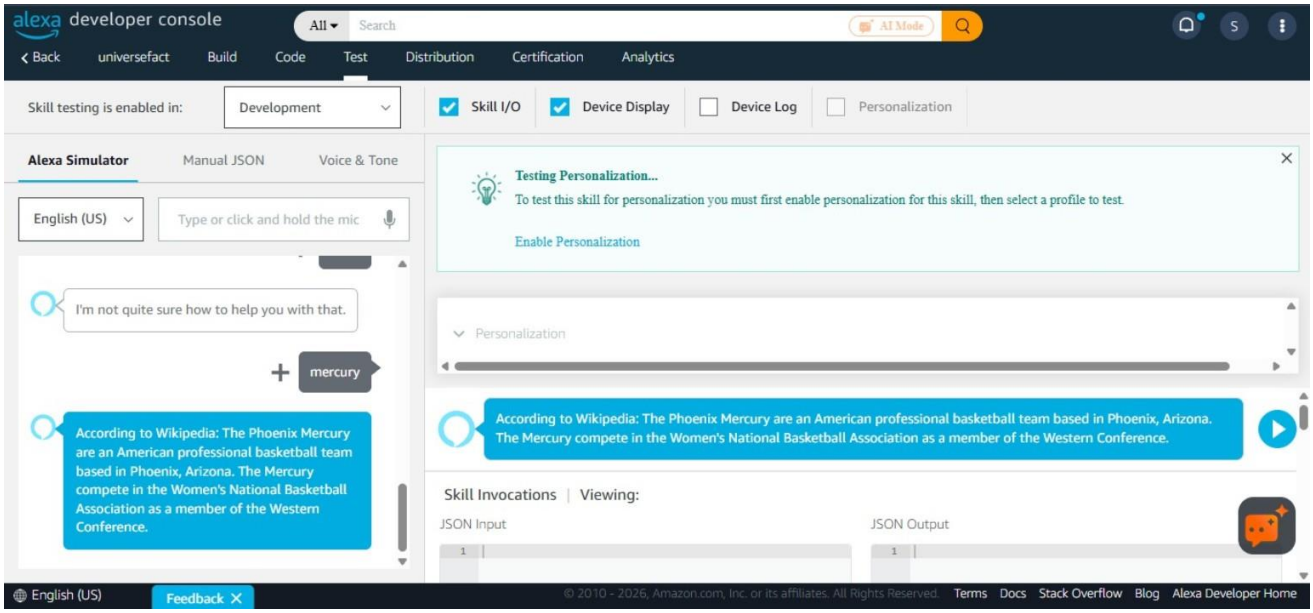


Figure 6: Alexa Developer Console

Amazon Bedrock Playground

Shows prompt-based generative AI testing using the Bedrock Playground.

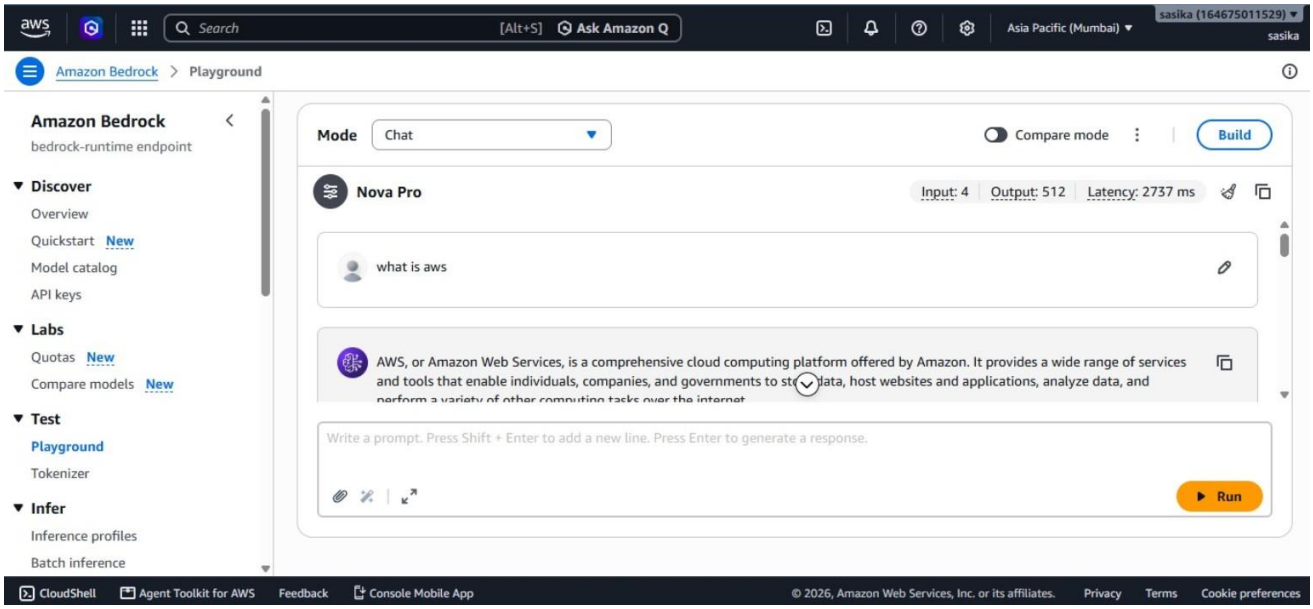


Figure 7: Amazon Bedrock Playground

Amazon Rekognition – Text Detection

Shows text detected from an image, including words such as FIND, BALANCE and IN LIFE.

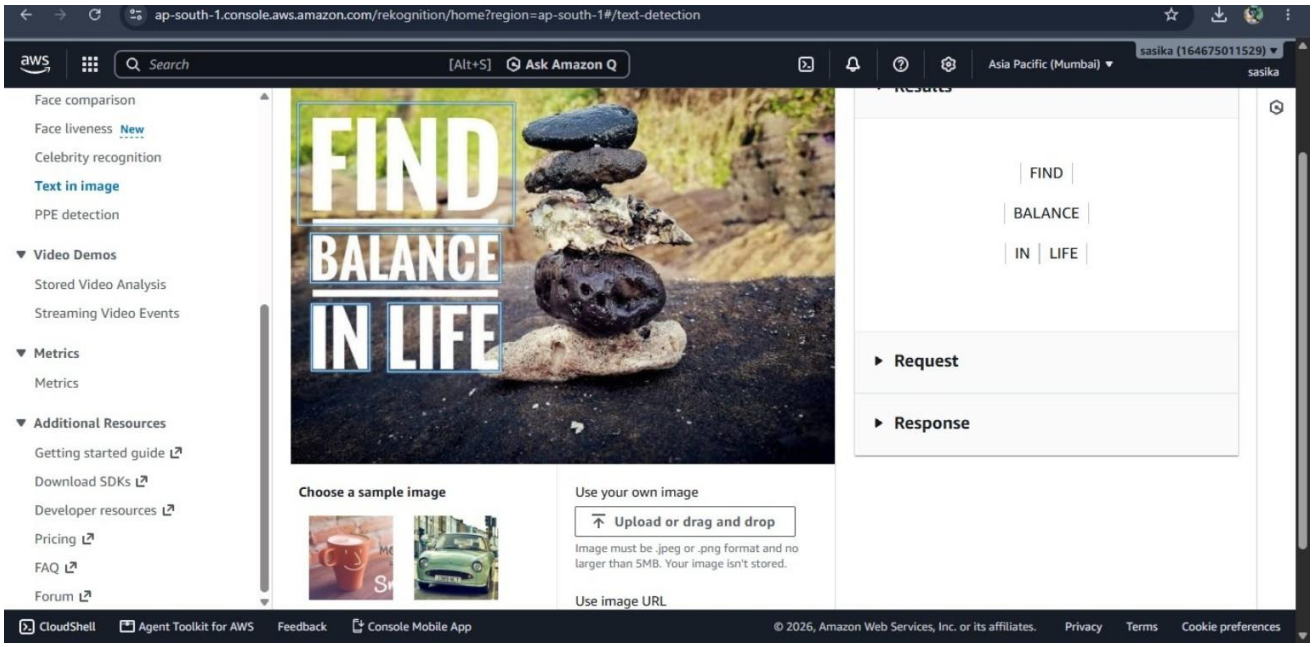


Figure 8: Amazon Rekognition – Text Detection

Amazon Comprehend – Key Phrase Analysis

Shows natural-language analysis and key phrases with confidence values.

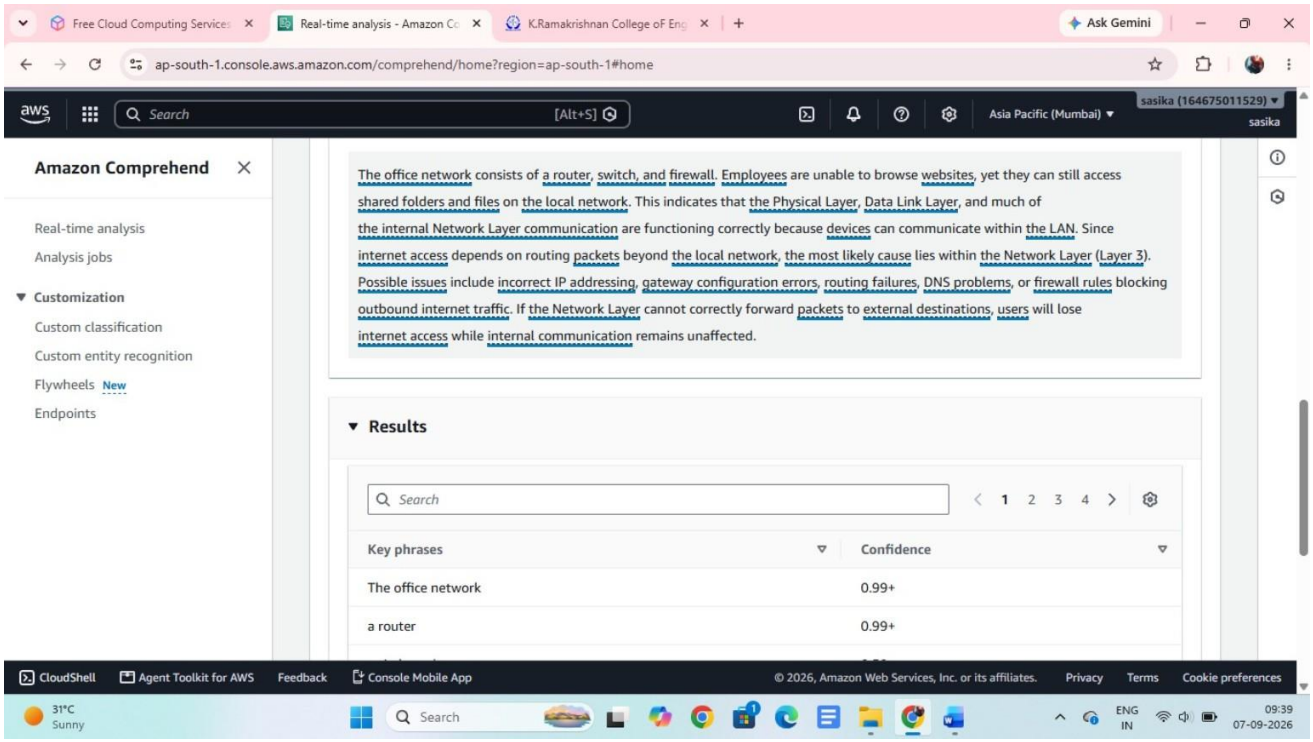


Figure 9: Amazon Comprehend – Key Phrase Analysis

Amazon Rekognition – Image Properties

Shows image properties including dominant colors and image-quality information.

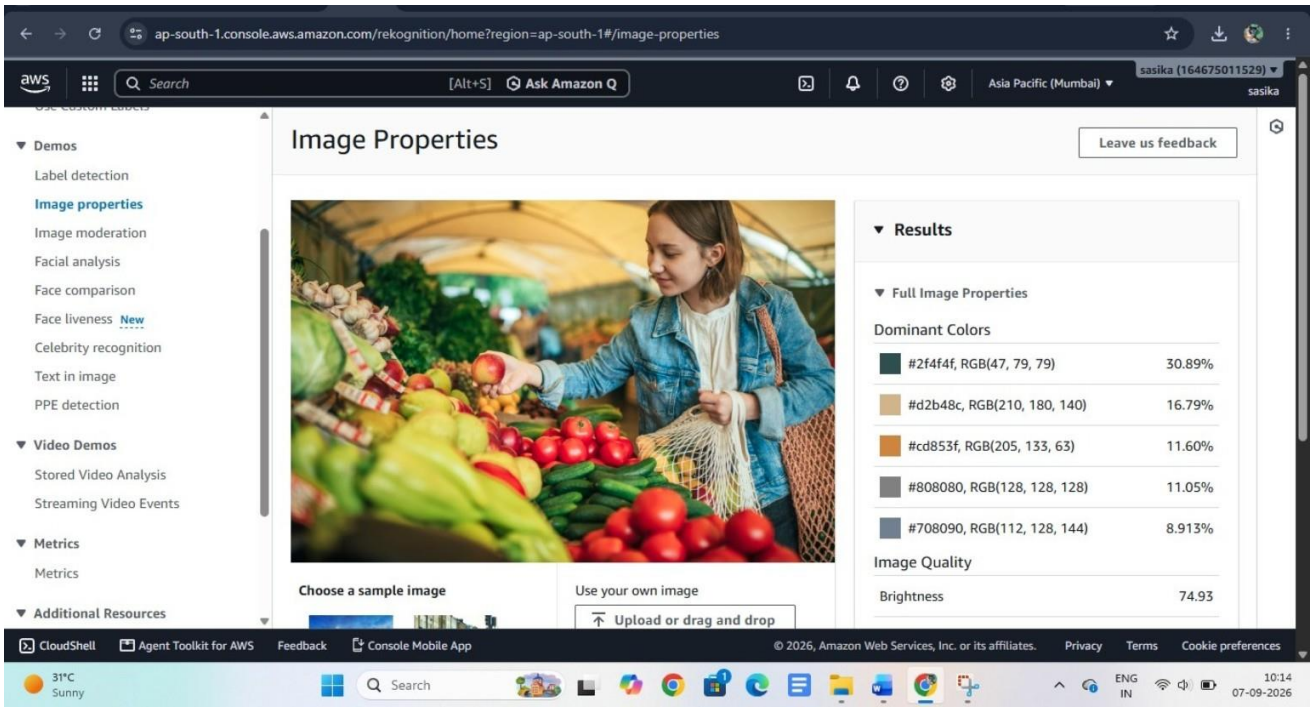


Figure 10: Amazon Rekognition – Image Properties

5. Key Learning Outcomes

The activities provide practical exposure to AWS infrastructure and AI capabilities, including EC2, S3, Athena, CloudFront, Bedrock, Alexa, Comprehend and Rekognition.

- Compute – Amazon EC2
- Storage – Amazon S3
- Analytics – Amazon Athena
- Content delivery – Amazon CloudFront
- Generative AI – Amazon Bedrock
- Voice/skill testing – Alexa Developer Console
- NLP – Amazon Comprehend
- Computer vision/OCR – Amazon Rekognition